Michael S. Albergo

Phone: US +1 (203) 738 9780 / UK +44 7712 899587 | Email: michaelsalbergo@gmail.com
Address: 99 Stonehedge Lane South, Guilford, CT 06437

EDUCATION

UNIVERSITY OF CAMBRIDGE, DOWNING COLLEGE — Cambridge, UK
MPhil in Physics, focus on machine learning — Expected Graduation Summer 2018

- Main project on implementing generative networks (GANs, VAEs) to simulate detector response for ATLAS detector at CERN.
- Secondary project with quantum information group to explore applicability of topological data analysis.
- Course attendances: quantum computing, quantum information theory
- I supervise the Part III masters course in theoretical particle physics, and I am the welfare officer for Downing graduate students.

HARVARD UNIVERSITY — Cambridge, MA
Bachelor of Arts in Physics: Recommended High Honors, cum laude — May 2017
Secondary in History (focus in 20th Century US)
Further Coursework in Computer Science, Statistics, Data Science, Mathematics, Machine Learning, Healthcare, and Biology

- Weissman Scholar
- Extracurriculars: Head Math Teaching Assistant (Linear Algebra, Multivariable Calculus), Commit 35 hours a week to Hasty Pudding Theatricals Musical, Co-founder of "science and politics" branch of Harvard Institute of Politics

GUILFORD HIGH SCHOOL — Guilford, CT

- Class President, President Spanish Honor Society, graduated with highest honors. ACT 35, Math SAT II 800 — June 2013

COMPUTER SKILLS AND PROJECTS

COMPUTER SCIENCE

- **Python, Matlab , C/C++, ROOT**
- Experience with/coursework in supervised and unsupervised learning: PCA, Bayesian stats, MDP, hidden markov models, neural networks, independent component analysis, canonical correlation analysis
- Maximizing Ebay's utility for iPhone sales: SVM and logistic regression to optimize sellers' profit maximizing strategy
- Diagnosing kidney inflammation through convolutional neural network techniques
- Creating predictor of photovoltaic output in energy transition for arbitrary molecule

RESEARCH EXPERIENCE

MASSACHUSETTS INSITITUTE OF TECHNOLOGY — Cambridge, MA
Adjunct Researcher, Cognitive Neuroscience — June 2017 – October 2017

- Dimensionality reduction of semantic space for understanding intrinsic neural sources of language processing in fMRI data.

EUROPEAN CENTER FOR NUCLEAR RESEARCH (CERN) — Geneva, Switzerland
Researcher, supersymmetry — July 2015 – July 2016

- Established kinematic variables to reject QCD background for s-top quark search in 0-Lepton and missing energy final state.
- Created algorithms for data quality of muon detection in the ATLAS experiment for real-time data acquisition

SLAC NATIONAL ACCELERATOR CENTER — Palo Alto, CA
SULI Researcher, plasma physics — June 2016 – August 2016

- Created X-ray propagation and phase retrieval simulation for imaging plasma materials to align theory with experiment.

YALE UNIVERSITY: EMERGENCY MEDICINE — New Haven, CT
Co-Investigator, medical imaging — June 2012 – June 2015

- Co-tailored a Convolutional Neural Network in Matlab to detect the presence and severity of hydronephrosis in ultrasound images to provide automated diagnosis or a validity check on physician decisions.
- Helped determine validity of bedside ultrasound in discerning presence of renal colic to avoid excessive radiation from CT scans.

LEADERSHIP

HARVARD UNIVERSITY MATHEMATICS DEPARTMENT — Cambridge, MA
Head Course Assistant (TA) — September 2014 – May 2017

- Head CA for multivariate calculus class. Teach a weekly section, verify homework solutions, and help designate responsibilities of other CAs. Increased teaching responsibilities for specific students.

HARVARD UNIVERSITY INSTITUTE OF POLITICS — Cambridge, MA
Co-founder of STEAM Initiative — May 2015 – June 2016

- Developed projects with companies and professors to help undergraduates explore the intersection of science and politics.
- Project examples: social media sentiment to beat polling predictions, case study analysis with ADS Ventures on biotech ethics.

MOVING TWICE USA — Cambridge, MA
US Co-Initiator — February 2014 – June 2015

- Moving Twice is an application that brings runners, charities and corporations together in such a way that for every mile a runner using the app runs, a sponsor company will donate a certain amount of money to a given charity.
- Innovated ways to market iPhone app service, acquire corporation sponsors. **Finalist**, Harvard i3 Innovation Competition.